Kalpesh Mahadik



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Education

B.Tech 2023 – 2027 **Vishwakarma Institute of Information Technology**, Artificial Intelligence and Data Science
- CGPA:8.59/10

Projects

Adaptive Mining Activity Monitoring using Sentinel 2.0 | ML, FastAPI, React

- Built end-to-end unsupervised ML pipeline for mining activity monitoring using Sentinel-2 time-series (Google Earth Engine).
- Implemented Isolation Forest anomaly detection achieving spectrally validated excavation signatures.
- Created React dashboard with FastAPI backend serving excavation maps, temporal profiles, and violation alerts. Github
- **Tech Used:** Python, Unsupervised Machine Learning, Isolation Forest, Google Earth Engine (GEE), Sentinel-2 Time-Series Data, FastAPI, React, REST APIs, Geospatial Analysis, Data Visualization

Drone-Based Crop Disease Detection System (Patented Granted)

- Designed and developed a drone-integrated computer vision system for real-time crop disease detection using Convolutional Neural Networks (CNNs).
- Built an end-to-end ML pipeline for image acquisition, preprocessing, model inference, and disease classification. Github
- Automated data storage and retrieval using Python, Flask, and MongoDB, enabling scalable and efficient dataset management.
- **Tech Stack:** Python, CNN, OpenCV, Flask, MongoDB, Drone Imaging, REST APIs

Marathi Movie Review Classification (NLP)

- Developed an NLP-based text classification system to categorize Marathi movie reviews by performing text preprocessing, tokenization, stemming, and lemmatization on regional language data.
- Extracted meaningful textual features and trained a machine learning classifier to analyze sentiment and review categories, gaining hands-on experience with regional language NLP challenges. Github
- **Tech Used:** Python, NLP, NLTK, Scikit-learn, TF-IDF, Text Preprocessing

E-commerce Sales & Business Intelligence Analytics (SQL | Python | Power BI)

- Performed end-to-end e-commerce data analysis using PostgreSQL (SQL) to derive insights on revenue growth, order frequency, customer behavior, and regional sales trends.
- Visualized insights using Python (Matplotlib) and designed an interactive Power BI dashboard with dynamic filters, drill-down capabilities, and DAX-based measures for real-time sales, profit, and performance analysis. Github
- **Tech Stack:** PostgreSQL, SQL, Python, Matplotlib, Power BI, DAX, Data Analysis, Business Intelligence, Data Visualization

Achievements

- **Most Innovative Solution Award – Aurora 2.0, IIT Dharwad:** Ranked most innovative solution among 250+ participants in a national-level hackathon.
- **Patent Granted – Drone-Based Crop Disease Detection System:** Patent for a CNN-based, drone-integrated real-time disease detection solution.
- **Data Structures Algorithms:** Solved 250+ questions on Leetcode.

Technologies

Programming Languages: Python, C++, SQL (MySQL, PostgreSQL)
Core Computer Science Concepts: Object-Oriented Programming (OOPs), Data Structures & Algorithms, DBMS, Problem Solving
Machine Learning & Data Science: TensorFlow, scikit-learn, NLTK, Pandas, NumPy, Matplotlib, Data Analysis, Data Visualization, Computer Vision
Tools & Platforms: Power BI, DAX, Jupyter Notebook, OpenCV, Git/GitHub